

periorbital wearable (patent pending)

thermopile (MLX90614)
ocular-surface temperature, 2 fields of view

camera + 940-nm LED
pupillography; IEC 62471-limited irradiance

4 Ag/AgCl dry electrodes per eye
local tetrapolar periorbital bioimpedance

temple PPG (MAX30102)
660 / 880 nm, 3 mm spacing

ESP32-S3 · I²C multiplexer · 250 mAh LiPo · BLE

CAD, poses
→
←
8 verdicts

physics digital twin (design loop)

3-D Pennes bioheat
122k-node FE, 14 tissues

complex-conductivity FE
lead fields, Geselowitz

Monte Carlo optics
PPG banana, LED absorption

pupil model (DDE)
autonomic gain, camera

verification & uncertainty
MMS order 1.8 · GCI < 2 % · Sobol, 8 parameters, 640 solves
ASME V&V 40 credibility framing

signal features

thermal
ambient-compensated ΔT

pupil
amplitude, latency, velocities

|Z|, phase; PPG
edema index, HRV, morphology

multimodal AI

fundus DR grader
ConvNeXt-T, ordinal 0–4

macular-oedema head
ordinal 0–2

systemic EHR model
boosted trees, n = 77,724

physiological branch
HRV, PPG, wearable features

ordinal late fusion
calibrated, threshold-optimised (QWK)

joint outputs
DR grade 0–4 · referable DR · DME · DN risk

evidence at submission (open data; credentialed mBRSET / BRSET pending)

DeepDRiD
1,589 images, 399 patients
patient-level 5-fold

IDRiD
516 images, external test
+ DME grades

APTOS 2019
3,662 images, India
second external test

Istanbul EHR
77,724 diabetic patients
DR + DN outcomes

PhysioNet ECG
94 subjects, HRV
DM vs control; DR null

PPG-BP
219 subjects
null + HTN control

mBRSET / BRSET
pending credentialing
(smartphone fundus + EHR)